

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

The client computer attempted to send DNS requests to the DNS server on UDP port 53, but the requests could not be delivered successfully.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

"udp port 53 unreachable"

The port noted in the error message is used for:

DNS (Domain Name System) services, which translate domain names such as www.yummyrecipesforme.com into IP addresses.

The most likely issue is:

The DNS server was not accepting requests on UDP port 53 because the DNS service was unavailable, misconfigured, or blocked by a firewall.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

13:24:32 (1:24 PM) according to the first timestamp in the tcpdump log.

Explain how the IT team became aware of the incident:

Multiple users reported that they could not access www.yummyrecipesforme.com and received a "destination port unreachable" error message. The analyst reproduced the issue and captured network traffic using tcpdump.

Explain the actions taken by the IT department to investigate the incident:

The analyst captured and reviewed network traffic logs, examined DNS requests sent from the client to the DNS server, and analyzed the ICMP responses returned by the server.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

- DNS requests were sent using UDP port 53.
- The DNS server IP address was 203.0.113.2.
- ICMP responses repeatedly returned "udp port 53 unreachable."
- The DNS server was unable to process DNS requests.
- Because DNS resolution failed, users could not access the website.

Note a likely cause of the incident:

A DNS service outage or network configuration issue caused UDP port 53 on the DNS server to become unavailable. Another possible cause is a firewall rule blocking DNS traffic to port 53.